Principal Component Analysis: Unifying Variance and Projection

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Abstract

Principal Component Analysis (PCA) is a fundamental technique for dimensionality reduction that identifies an optimal set of orthogonal features (the principal component directions) providing the best low-rank approximation of a dataset. This approximation can be characterized in two equivalent ways: minimizing reconstruction error (based on the Eckart–Young theorem), or maximizing variance in the feature space (based on the Rayleigh–Ritz theorem). The reconstruction error formulation involves computing the Singular Value Decomposition (SVD) of the data matrix, while the variance-maximization viewpoint requires diagonalizing the covariance matrix of the data. In this paper, we establish the correctness of low-rank approximations through a proof of the Eckart-Young theorem, and present detailed explanations of both the SVD-based and covariance-based procedures. We then prove that the two approaches yield identical principal subspaces and conclude with a comparison of their computational costs and efficiency in a practical setting.

1 Introduction

Datasets in today's world often contain millions of features—the average number of features in many medical and financial datasets ranges from 21–50 [1]. High dimensionality (a large number of features) increases the distance between individual data points, leading to increased "scatter" within the data; this makes it difficult to fit regression models for statistical analyses. Datasets can be viewed as matrices with feature columns and row entries. A high-dimensional dataset corresponds to a high-rank matrix with several linearly independent feature columns; such high-rank matrices are computationally expensive to store and manipulate.

Dimensionality reduction therefore plays a central role in modern data analysis and machine learning. Principal Component Analysis (PCA) is a widely used dimensionality-reduction technique that computes the best low-rank approximation of a data matrix. Intuitively, PCA provides a faithful "summary" of the dataset along a smaller set of orthogonal, uncorrelated features (the principal component directions) which capture the greatest variance in the data. Equivalently, orthogonally projecting a data matrix onto the subspace spanned by the principal component directions yields a low-rank approximation that is "closest" to the original matrix in the Frobenius-norm sense.

The earliest formulation of PCA, due to Karl Pearson in 1902, relied solely on minimizing orthogonal projection error to determine the principal component directions. In 1936, Eckart and Young strengthened this viewpoint by proving that the SVD-based orthogonal-projection approach indeed produces the low-rank approximation that minimizes reconstruction error (as measured by the Frobenius norm). More recent developments in probability theory provide an alternative formulation of PCA based on identifying the directions of maximum variance in a dataset. While both the variance-maximization and SVD-based formulations yield the same principal subspace, the SVD approach typically requires fewer operations and is therefore more computationally efficient.

In this paper, we first develop the projection-based viewpoint of principal components. This view of PCA formulates the principal component directions of a matrix as the vectors which generate a subspace that minimizes the reconstruction error of a projected approximation matrix. We provide a detailed proof of the Eckart–Young–Mirsky Theorem, which states that the truncated SVD of a data matrix minimizes total reconstruction error. We also present an alternative probabilistic framework for PCA which is widely used in statistical analyses. In this framework, the principal component directions are defined as the orthogonal

(uncorrelated) vectors along which the variance of the data is maximized. We discuss why these directions of maximal variance coincide with the eigenvectors of the covariance matrix of a centered data matrix.

The first section introduces low-rank matrix approximations and the reconstruction error of such approximations; here we introduce the optimization problem underlying PCA (minimizing the total squared reconstruction error). This is followed by a detailed overview of the "PCA via SVD" algorithm. In the next section on maximizing variance, we introduce the notion of variance along a direction within our data, connecting this to covariance matrices. The Rayleigh–Ritz theorem is used to show that the variance is maximized along the eigenvectors of the covariance matrix, thereby identifying these eigenvectors as the principal component directions. In the last section, we show that the variance-maximization method and SVD-based approach to PCA yield the same principal subspace. We conclude the paper by comparing the computational efficiency and costs of both frameworks.

2 PCA via Orthogonal Projections

Suppose we have a $p \times n$ data matrix X, where each $j-$th row of X $(1 \leq j \leq p)$ is a data vector with n features (columns): $x_j = (x_j^{(1)}, ..., x_j^{(n)})$. The data matrix X is said to be "high-rank" if $\text{rank}(X) \approx n$.

Let $k \in \mathbb{N}$ with $k < \text{rank}(X) \leq n$. PCA allows us find *the best rank$-k$ approximation of* X, denoted by X_k. Intuitively, PCA allows us to form an "accurate" summary of our dataset, by finding the "best" set of k new orthogonal features and re-expressing our data in this smaller basis. But what do we mean by the "best" rank$-k$ approximation?

2.1 Minimizing Reconstruction Error

Let U_k be a $k-$dimensional subspace of our *feature space*: $\text{Col}(X^T) \subseteq \mathbb{R}^n$, spanned by the orthonormal vectors: $\{u_1, ..., u_k\}$ $(u_k \in \mathbb{R}^n)$; we write $[U_k]$ to denote the matrix formed by joining these basis vectors : $[u_1|, ..., |u_k]$. Let $P_k = [U_k][U_k]^T$ be the $n \times n$ orthogonal projection matrix of U_k, and x_j be a data vector (row) in X.
We approximate a data vector x_j in X by its orthogonal projection onto U_k:

$$\hat{x}_j := x_j P_k \ .$$

Now, the squared error of our projected approximation of x_j is $||\hat{x}_j - x_j||^2$. This is called the *reconstruction error* of $\hat{x}_j$.

Definition 2.1 (*Total Reconstruction Error*)**.** For a given subspace U_k, and a centered data matrix $X = (x_1, x_2, ..., x_p)^T$, the total reconstruction error $E(U_k)$ is the sum of squared distances between each data vector x_j and its projected approximation $\hat{x}_j$:

$$E(U_k) = \sum_{j=1}^{p} ||x_j - \hat{x}_j||^2 \ .$$

The goal of PCA is to find the set of vectors $\{u_1, ..., u_k\}$ to span U_k, which minimizes $E(U_k)$; these vectors are called the ***principal component directions***, or ***PC vectors*** of X.

Expressing $E(U_k)$ in terms of matrices reveals that finding principal component directions lets us compute X_k, the closest rank$-k$ matrix approximation to X. The Frobenius norm of a matrix makes this notion of closeness precise.

Definition 2.2 (*Frobenius Norm*)**.** Let A be an $p \times n$ matrix. The *Frobenius Norm* of A, denoted $||A||_F$, is given by:

$$||A||_F = \sqrt{\sum_{i=1}^{p} \sum_{j=1}^{n} A_{ij}^2} = \sqrt{\text{Trace}(A^T A)} \ .$$

We define the **rank$-k$ approximation matrix ($\hat{X}$)** of projected vectors by $\hat{X} := (\hat{x_1}, ..., \hat{x_p})^T = XP_k$. Thus the total reconstruction error is:

$$E(U_k) = \sum_{j=1}^{p} ||x_j - \hat{x_j}||^2 = ||X - \hat{X}||_F^2 \ .$$

PCA finds X_k, the rank$-k$ approximation matrix minimizing total reconstruction error:

$$X_k := \underset{\text{rank}(\hat{X}) \leq k,\ \hat{X}_{p \times n}}{\arg\min} ||X - \hat{X}||_F^2$$

Computing X_k requires the Singular Value Decomposition of our data matrix: $X = R\Sigma V^T$. We select the first k columns in V (these are our PC vectors) to span the dim$-k$ subspace U_k; then projecting X onto U_k provides X_k. We will walk through the entire process with an example.

2.2 PCA via SVD: The Procedure

Theorem 1 (Singular Value Decomposition of a Matrix)**.** *Let X be a $p \times n$ matrix, with rank(X) $= r$. Then X can be factorized in the following form:*

$$X = R\Sigma V^T$$

where $R_{p \times p}$ and $V_{n \times n}$ are orthogonal matrices. The columns of V are precisely the eigenvectors $v_1, ..., v_n$ of $X^T X$, and the r singular values $\sigma_1 \geq ... \geq \sigma_r$ of X are given by $\sigma_i = \sqrt{\lambda_i}$, where λ_i is the eigenvalue corresponding to v_i in V for $i = 1, ..., r$.
The columns of R are the eigenvectors of XX^T, and Σ is a pseudo-diagonal $p \times n$ matrix filled with the singular values :

$$\Sigma = \begin{bmatrix} \text{diag}(\sigma_1, \ldots, \sigma_r) & 0 \\ 0 & 0 \end{bmatrix}.$$

Since XX^T and $X^T X$ are symmetric and positive-semidefinite, the Spectral Theorem guarantees the existence of a full basis of orthonormal eigenvectors for XX^T and $X^T X$ respectively, correspondent to real and non-negative eigenvalues. Thus each singular value: $\sigma_k = \sqrt{\lambda_k}$ and the matrices R, V are always well-defined; every matrix can be factorized into this convenient form. In the following example, we use the SVD of a matrix to find its rank-2 approximation.

Example 1. Consider the 6×3 data matrix:

$$X = \begin{bmatrix} 3 & 2 & 1 \\ -3 & 2 & -1 \\ 3 & -2 & 0 \\ -3 & -2 & 0 \\ 3 & 0 & -1 \\ -3 & 0 & 1 \end{bmatrix}.$$

The columns of X are already centered (the mean of each column is zero), and rank$(X) = 3$. Our goal is to compute its rank-2 approximation, X_2.
If the SVD of X is given by $R\Sigma V^T$, where $R_{6\times6}$ and $V_{3\times3} = [v_1, v_2, v_3]$ are orthogonal matrices, then our first two PC vectors are stored in $V_2 := [v_1, v_2]$. Immediately, our projection matrix P_2 will be $V_2 V_2^T$; then we can compute X_2 by $X_2 = XP_2$. Here, $X^T X$ is given by:

$$X^T X = \begin{bmatrix} 54 & 0 & 0 \\ 0 & 16 & 0 \\ 0 & 0 & 4 \end{bmatrix}.$$

$X^T X$ is already diagonal, so its eigenvalues are $54, 16$ and 4 corresponding to the eigenvectors $e_1, e_2, e_3 \in \mathbb{R}^3$. Thus, our first two singular values are $\sigma_1 = \sqrt{54} = 3\sqrt{6}$, $\sigma_2 = \sqrt{16} = 4$, and $V = [\ e_1 \mid e_2 \mid e_3\ ]$. From

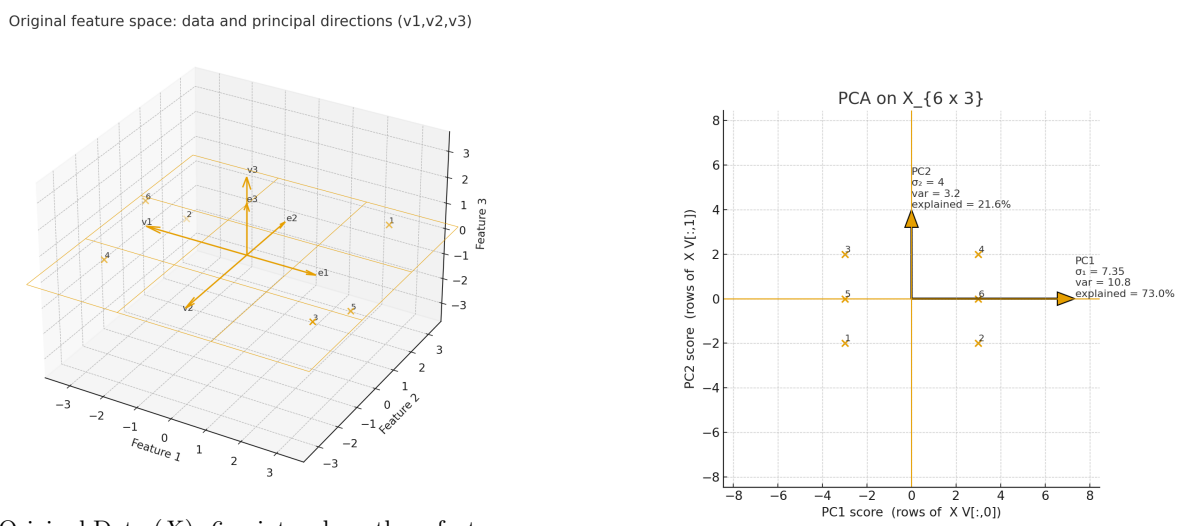
here, we form $V_2 = [\ e_1 \ \mid \ e_2\ ]$, whereby $P_2 = V_2 V_2^T$:

$$P_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus P_2 projects vectors in $\mathbb{R}^3$ onto the subspace $\mathbb{R}^2$; now our best rank-2 approximation X_2 of X is simply:

$$X_2 := XP_2 = \begin{bmatrix} 3 & 2 & 0 \\ -3 & 2 & 0 \\ 3 & -2 & 0 \\ -3 & -2 & 0 \\ 3 & 0 & 0 \\ -3 & 0 & 0 \end{bmatrix}.$$

Equivalently, we could have used the truncated SVD of X, keeping only the first two columns of R, Σ, V for our rank-2 approximation: $X_2 = R_2 \Sigma_2 V_2^T = XP_2$. In Figure 1, we see a visualization of our approximation: the 6 data points plotted along the 3 dimensions from our original rank-3 dataset are projected onto the principal subspace $\mathbb{R}^2$, offering a rank-2 approximation along the first two principal component directions.



(a) Original Data (X): 6 points, along three features (columns)

(b) Projected data (X_2), along first two PC vectors

Figure 1: PCA via orthogonal projections

3 The Eckart - Young Theorem

In our earlier example, the result X_2 matched our intuition for approximating X. But can we guarantee that PCA always works, for *any* matrix? The Eckart-Young Theorem promises that we can: the truncated SVD of our data matrix always offers its best low rank approximation.

Theorem 2 (The Eckart - Young Theorem)**.** *Let $X \in \mathbb{R}^{p \times n}$ of rank(X) $= r$ have the SVD:*

$$X = R \Sigma V^\top$$

where $R \in \mathbb{R}^{p \times p}$ and $V \in \mathbb{R}^{n \times n}$ are orthogonal, and $\Sigma \in \mathbb{R}^{p \times n}$ is pseudo-diagonal containing singular values

$$\sigma_1 \geq \cdots \geq \sigma_r > 0$$

Let $V_k = [v_1|\cdots|v_k] \in \mathbb{R}^{n\times k}$, $R_k = [r_1|\cdots|r_k] \in \mathbb{R}^{p\times k}$, and $\Sigma_k = \text{diag}(\sigma_1, \ldots, \sigma_k)$. Then, $X_k = R_k\Sigma_k V_k^T$ is a rank$-k$ matrix satisfying:

$$||X - X_k||_F^2 \leq ||X - Y||_F^2$$

for any $Y \in \mathbb{R}^{p\times n}$ with rank$(Y) \leq k$

The proof requires two simple results.

Lemma 3. Let A be any matrix, and Q, W are orthogonal matrices of compatible dimensions. Then, $||QAW||_F^2 = ||A||_F^2$

Proof. $||QAW||_F^2 = \text{Tr}((QAW)^T(QAW)) = \text{Tr}(W^TA^TQ^TQAW) = \text{Tr}(W^TA^TAW)$. Since the trace is cyclic, $\text{Tr}(W^TA^TAW) = \text{Tr}(WW^TA^TA) = \text{Tr}(A^TA) = ||A||_F^2$. □

Corollary 3.1. In Theorem 2,

$$||X - X_k||_F^2 = \sum_{i=k+1}^{r} \sigma_i^2$$

Proof. We have $X = R\Sigma V^T$ where $\Sigma = [\text{diag}(\sigma_1, ..., \sigma_r), 0, ..., 0]_{p\times n}$, and $X_k = R_k\Sigma_k V_k^T$. Let Σ_k^* be the matrix:

$$\Sigma_k^* = \begin{bmatrix} \text{diag}(\sigma_1, \ldots, \sigma_k) & 0 \\ 0 & 0 \end{bmatrix}_{p\times n}$$

Then $X_k = R_k\Sigma_k V_k^T = R\Sigma_k^* V^T$; so $||X - X_k||_F^2 = ||R(\Sigma - \Sigma_k^*)V^T||_F^2 = ||\Sigma - \Sigma_k^*||_F^2$. Thus,

$$||X - X_k||_F^2 = ||\Sigma - \Sigma_k^*||_F^2 = \sum_{i=1}^{p}\sum_{j=1}^{n}(\Sigma_{ij} - (\Sigma_k^*)_{ij})^2 = \sum_{i=k+1}^{r} \sigma_i^2$$

□

Lemma 4. Multiplication by an orthogonal matrix preserves rank.

Proof. This is true of multiplication by any invertible matrix; for any $A_{p\times n}$ and orthogonal $P_{p\times p}$, $\text{Ker}(PA) = \{x : PAx = 0\} = \{x : Ax = 0\}$; hence, $\dim(\text{Ker}(PA)) = \dim(\text{Ker}(A))$. By the Rank-Nullity theorem, $\text{rank}(PA) = \text{rank}(A)$. Similarly, $\text{rank}(AQ) = \text{rank}(A)$ for any orthogonal $Q_{n\times n}$. □

We are ready for the main proof of **Theorem 2**.

Proof. Let $Y_{p\times n}$ with $\text{rank}(Y) \leq k$ be any candidate for the minimizer, and R, V be orthogonal matrices with $X = R\Sigma V^T$ as in Theorem 2. Define the matrix $Z := R^TYV$, so $Y = RZV^T$. By Lemma 4, $\text{rank}(Z) = \text{rank}(Y) \leq k$. By Lemma 3,

$$||X - Y||_F^2 = ||R(\Sigma - Z)V^T||_F^2 = ||\Sigma - Z||_F^2$$

Thus, minimizing $||X - Y||_F^2$ over $\text{rank}(Y) \leq k$ is equivalent to minimizing $||\Sigma - Z||_F^2$ over $\text{rank}(Z) \leq k$. Note that in Σ, $\Sigma_{ij} = 0$ for any $i \neq j$ or $j > r$. Thus for any matrix $Z_{p\times n}$, we have:

$$E(Z) = ||\Sigma - Z||_F^2 = \sum_{j=1}^{n}\sum_{i=1}^{p}(\Sigma_{ij} - Z_{ij})^2 = \sum_{i=1}^{r}(\Sigma_{ii} - Z_{ii})^2 + \sum_{i\neq j,\ i>r} Z_{ij}^2 + \sum_{i>r} Z_{ii}^2$$

Any nonzero off-diagonal entries in Z will *increase* $E(Z)$ by $\sum_{i\neq j,\ i>r} Z_{ij}^2$, so any minimizer Z must be diagonal; since $\text{rank}(Z) \leq k$, there can be at most k nonzero diagonal entries entries in Z. Then, $E(Z) = \sum_{i=1}^{r}(\sigma_i - Z_{ii})^2 + \sum_{i>r} Z_{ii}^2$ is minimized by setting $Z_{ii} = \sigma_i$ for the largest k singular values: $\sigma_1 \geq ... \geq \sigma_k$; thus Σ_k^* in Corollary 3.1 satisfies:

$$||\Sigma - \Sigma_k^*||_F^2 \leq ||\Sigma - Z||_F^2 \text{ , for any } Z_{p\times n} \text{ with } \text{rank}(Z) \leq k.$$

Letting $X_k = R\Sigma_k^* V^T = R_k\Sigma_k V_k^T$, we have:

$$||X - X_k||_F^2 \leq ||X - Y||_F^2$$

for any $Y_{p\times n}$ with $\text{rank}(Y) \leq k$. □

4 Maximizing Variance, Minimizing Error

In performing any kind of analysis on a dataset, it is natural to ask: which features best explain the variance in our dataset? The **variance-maximization** viewpoint of PCA finds directions in our feature space that are mutually orthogonal and maximize variance, providing a non-redundant set of features that best "explains" our data. In this section, we see how PCA can be performed via the Rayleigh-Ritz variance-maximization approach, and prove that this approach yields exactly the same principal subspace as our SVD-approach for any centered data matrix.

4.1 Centering the Data and the Sample Covariance Matrix

Recall our $p \times n$ data matrix: $X = (x_1, ..., x_p)^T$ where each row $x_j \in \mathbb{R}^n$ is a data vector with n features. We define the **mean-vector** of the rows of X as $\mu := \frac{1}{p}\sum_{j=1}^{p} x_j \in \mathbb{R}^n$. Then we define the *centered data matrix* X_c by subtracting this mean-vector from each row:

$$X_c := \begin{bmatrix} x_1 - \mu \\ \vdots \\ x_p - \mu \end{bmatrix} \in \mathbb{R}^{p \times n}.$$

Thus, the mean of the rows of X_c is the zero vector in $\mathbb{R}^n$.

Definition 4.1 (Sample covariance matrix)**.** The *sample covariance matrix* of X is the $n \times n$ matrix

$$S := \frac{1}{p-1} X_c^\top X_c \in \mathbb{R}^{n \times n}.$$

The (i, i)–entry of S is the empirical variance of the i-th feature across the p data points, and the off-diagonal entry S_{ik} is the empirical covariance between features i and k. Intuitively, S_{ik} tells us how strongly the $i-$th and $k-$th features influence each other's values. Since S is of the form $S = \frac{1}{p-1} A^\top A$ with $A = X_c$, it is symmetric and positive semi-definite.

4.2 Variance in a Direction

The variance-maximization viewpoint of PCA is formulated in terms of the empirical variance of the data when projected onto a given direction in feature space.

Let $u \in \mathbb{R}^n$ be a unit vector $||u|| = 1$. We interpret u as *a direction in the n-dimensional feature space.* For each centered data vector $x_j^c := x_j - \mu \in \mathbb{R}^n$, we define its scalar projection onto u by:

$$y_j := x_j^c u^\top \in \mathbb{R}, \qquad j = 1, \ldots, p.$$

Collecting these into a vector, we define:

$$Y := X_c u^\top \in \mathbb{R}^{p \times 1},$$

whose j-th entry is y_j. The empirical variance of this one-dimensional dataset $\{y_1, \ldots, y_p\}$ is

$$\mathrm{Var}(u) := \frac{1}{p-1} \sum_{j=1}^{p} y_j^2 = \frac{1}{p-1} Y^\top Y.$$

Using the definition of Y and the covariance matrix S, we compute

$$\mathrm{Var}(u) = \frac{1}{p-1} u X_c^\top X_c u^\top = u S u^\top.$$

Definition 4.2 (Directional variance)**.** Let $u \in \mathbb{R}^n$ be a unit vector. The *variance of the data in the direction* u is defined by

$$\mathrm{Var}(u) := u S u^\top.$$

Thus, to say that a direction u has "large variance" means that the one-dimensional dataset of projections $\{x_j^c u^\top\}_{j=1}^p$ has large empirical variance, as measured by $u S u^\top$.

4.3 Principal Components as Variance-Maximizing Directions

We now formulate PCA as the problem of finding orthonormal directions in feature space that maximize the directional variance.

The first principal component

The first principal component direction is defined as the unit vector v_1 that maximizes the variance $uSu^\top$ among all unit vectors $u \in \mathbb{R}^n$. We consider the optimization problem

$$\begin{aligned} \text{maximize} \quad & uSu^\top \\ \text{subject to} \quad & \|u\| = 1. \end{aligned} \tag{1}$$

The objective $uSu^\top$ is called the *Rayleigh quotient* of S at u. The following result characterizes the maximizers of (1).

Theorem 5 (Rayleigh–Ritz characterization)**.** *Let $S \in \mathbb{R}^{n\times n}$ be symmetric, with eigenvalues*

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0,$$

and an orthonormal basis of eigenvectors $v_1, \ldots, v_n \in \mathbb{R}^n$, so that

$$Sv_k = \lambda_k v_k, \qquad k = 1, \ldots, n.$$

Then

$$\max_{\|u\|=1} uSu^\top = \lambda_1,$$

and every maximizer u is an eigenvector corresponding to λ_1, i.e., $u = \pm v_1$.

A standard proof proceeds via Lagrange multipliers. Define the Lagrangian

$$L(u, \alpha) := uSu^\top - \alpha(u^\top u - 1),$$

and note that critical points satisfy

$$\nabla_u L(u, \alpha) = 0 \quad \iff \quad 2Su - 2\alpha u = 0 \quad \iff \quad Su = \alpha u,$$

so that u must be an eigenvector of S; then the maximum is attained at an eigenvector corresponding to the largest eigenvalue λ_1.

Definition 4.3 (First principal component direction)**.** The *first principal component direction* $v_1 \in \mathbb{R}^n$ is any unit eigenvector of S corresponding to its largest eigenvalue λ_1. By Theorem 5, v_1 maximizes the directional variance:

$$\mathrm{Var}(v_1) = v_1 S v_1^\top = \lambda_1 = \max_{\|u\|=1} uSu^\top.$$

Iteratively finding the PC Directions

The second principal component direction is defined as the unit vector v_2 that is orthogonal to v_1 and maximizes the directional variance. This leads to the constrained optimization problem

$$\begin{aligned} \text{maximize} \quad & uSu^\top \\ \text{subject to} \quad & \|u\| = 1, \\ & u^\top v_1 = 0. \end{aligned} \tag{2}$$

By restricting the Rayleigh quotient to the subspace *orthogonal* to v_1, we find that the maximizer is an eigenvector corresponding to the second largest eigenvalue λ_2, and so on.

Theorem 6 (*Principal components as eigenvectors of the covariance matrix*)**.** *Let* $S = \frac{1}{p-1} X_c^\top X_c$ *be the sample covariance matrix of the centered data* X_c*, and let*

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0$$

be its eigenvalues with corresponding orthonormal eigenvectors $v_1, \ldots, v_n$*. Then:*

1. *For each* $k \in \{1, \ldots, n\}$*, the* k*-th principal component direction* v_k *is an eigenvector of* S *corresponding to the* k*-th largest eigenvalue* λ_k*.*
2. *The variance of the data in direction* v_k *is*

$$\mathrm{Var}(v_k) = v_k S v_k^\top = \lambda_k.$$

If we collect the first k principal directions into the matrix

$$V_k := [v_1 \mid \cdots \mid v_k] \in \mathbb{R}^{n \times k},$$

then the orthogonal projector onto the k-dimensional principal subspace is $P_k = V_k V_k^\top$. The matrix of *principal component scores* (the coordinates of the centered data in this principal subspace) is

$$Y := X_c V_k \in \mathbb{R}^{p \times k},$$

where the j-th row of Y consists of the k principal component scores of x_j.

Example 2. Recall our earlier example of computing the rank-2 approximation of the 6×3 matrix X in Example 1. The covariance matrix of X is $S = \frac{1}{p-1} X^T X = \frac{1}{5} X^T X$, where p is the number of rows in X. By Theorem 6, the first two eigenvectors of S are the first two principal component directions $v_1,\ v_2$ of X. Let us compute S and $v_1,\ v_2$:

$$S = \begin{bmatrix} \frac{54}{5} & 0 & 0 \\ 0 & \frac{16}{5} & 0 \\ 0 & 0 & \frac{4}{5} \end{bmatrix} \implies v_1 = (1,0,0)^T \text{ with } \lambda_1 = \frac{54}{5}, \quad v_2 = (0,1,0)^T \text{ with } \lambda_2 = \frac{16}{5}$$

Then with $V_2 = [v_1 | v_2]$, we find the *same* principal subspace (along which the variance in our data across two dimensions is maximized) as in Example 1. Again, our projection matrix is $P_2 := V_2 V_2^T$, which provides the 2-dimensional approximation for X:

$$\hat{X} = X P_2 = \begin{bmatrix} 3 & 2 & 1 \\ -3 & 2 & -1 \\ 3 & -2 & 0 \\ -3 & -2 & 0 \\ 3 & 0 & -1 \\ -3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 0 \\ -3 & 2 & 0 \\ 3 & -2 & 0 \\ -3 & -2 & 0 \\ 3 & 0 & 0 \\ -3 & 0 & 0 \end{bmatrix}.$$

The fact that this approximation matches X_2 in Example 1 is no coincidence; in the following section, we rigorously justify why the directions of maximal variance in our data span the same principal subspace as the directions along which reconstruction error is minimized.

5 Equivalence of Both Approaches

Now we connect the variance-maximization viewpoint developed above to the SVD formulation of PCA introduced earlier. Let $X_c = R \Sigma V^\top$ be the SVD of the centered data matrix, where

$$R \in \mathbb{R}^{p \times p}, \quad V \in \mathbb{R}^{n \times n} \quad \text{are orthogonal,}$$

and $\Sigma \in \mathbb{R}^{p \times n}$ is pseudo-diagonal with diagonal entries

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0,$$

where $r = \text{rank}(X_c)$, and all other entries of Σ are zero. Then:

$$X_c^\top X_c = V\Sigma^\top R^\top R\Sigma V^\top = V\Sigma^\top \Sigma V^\top = V\Sigma^2 V^\top,$$

where Σ^2 denotes the $n \times n$ diagonal matrix whose diagonal entries are $\sigma_1^2, \ldots, \sigma_r^2$, followed by zeros. Consequently, the covariance matrix can be written as

$$S = \frac{1}{p-1} X_c^\top X_c = V\left(\frac{1}{p-1}\Sigma^2\right)V^\top.$$

From this expression, we see that the eigenvectors of S are precisely the columns of V, i.e. the right singular vectors of X_c, and the eigenvalues of S are given by $\lambda_i = \frac{\sigma_i^2}{p-1},\ i = 1, \ldots, r$, with remaining eigenvalues equal to zero. Thus, the variance-maximization principal directions obtained as eigenvectors of S coincide with the principal component directions obtained from the SVD of X_c. In particular, if we define

$$V_k := [v_1 \mid \cdots \mid v_k]$$

as in Theorem 6, then V_k is exactly the matrix formed by the first k columns of V in the SVD. The orthogonal projector $P_k = V_k V_k^\top$ and the rank-k approximation

$$X_{c,k} := X_c P_k = R_k \Sigma_k V_k^\top$$

are therefore identical in both viewpoints: PCA via best rank-k approximation (Eckart–Young) and PCA via variance maximization produce the same k-dimensional approximation of a centered data matrix.

5.1 Computational Costs

Even though PCA via variance maximization and PCA via low-rank approximation are mathematically equivalent, the algorithms associated with these two formulations have different operation counts and storage costs. In the variance-maximization approach, we work with the sample covariance matrix

$$S = \frac{1}{p-1} X_c^\top X_c \in \mathbb{R}^{n\times n},$$

where $X_c \in \mathbb{R}^{p\times n}$ is the centered data matrix with p data points and n features. Forming S requires multiplying an $n \times p$ matrix with a $p \times n$ matrix, which has arithmetic cost on the order of $O(pn^2)$.
Once S is formed, computing the principal component directions requires finding the eigen-decomposition of S, which is an $n \times n$ matrix; the cost of this step is $O(n^3)$ when the matrix is *dense* (with several nonzero entries). Thus, using the covariance matrix involves both: the cost of forming S, and the cost of its eigen-decomposition; we also incur storage costs associated with an $n \times n$ matrix.

In the SVD-based formulation, we instead work directly with the centered data matrix X_c and compute its singular value decomposition

$$X_c = R\Sigma V^\top, \qquad R \in \mathbb{R}^{p\times r},\ \Sigma \in \mathbb{R}^{r\times r},\ V \in \mathbb{R}^{n\times r},$$

where $r = \text{rank}(X_c)$. Standard dense algorithms for the full SVD of a $p \times n$ matrix have arithmetic cost on the order of $\min\{O(pn^2), O(p^2n)\}$, depending on whether $p \geq n$ or $n \geq p$. When only the first $k \ll \min\{p, n\}$ principal components are required, truncated or iterative SVD methods can be used, whose cost typically scales like $O(pnk)$ rather than that of a full decomposition.

Thus, in high-dimensional settings, SVD-based methods prove preferable due to quadratic operation counts. There are certain special cases of *sparse* matrices (with mostly zero-entries) where the eigen-decomposition of the covariance matrix can be computed more efficiently through a QR decomposition. Therefore, the SVD-based method is more frequently employed on dense datasets, while the variance-based approach is better suited to sparse datasets.

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